

An executive brief on market, opportunity, and the case for agentic AI.

Prepared for Al Weiss

By Brian Mulhall · Chief Executive Officer · CS Analytical

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One sentence.

Container Closure Integrity is the single most regulated, most ignored, and most under-served line item on a sterile-drug spec sheet.

CS Analytical exists to fix that — and to scale the fix with agentic AI partners.

Brian Mulhall — Chief Executive Officer

Thirty years building the labs the industry actually trusts.

1990s	Late 1990s	2002 — 2015	2015 — today
Sales, marketing, and national management at Organon (Schering Plough) and Ferring.	VP, Pharmaceutical Services, SGS US Testing — chemistry, microbiology, and toxicology across the U.S. and Canada.	Founded Whitehouse Analytical Laboratories. Built to 48 FTE, 25,000 sq ft. Sold in 2015.	CEO, Leak Detection Associates. Advisor and BD lead, Visikol. Now CEO, CS Analytical.

Architect of the first FDA-regulated, cGMP Container Closure Integrity Testing laboratory in the world — the work that informed USP <1207>.

CS Analytical Laboratory

A center of excellence, not a generalist lab.

Specialty depth

CCIT done at a level the big labs can't staff for — helium leak, HVLD, vacuum decay, dye ingress, methods development, validation.

Regulatory posture

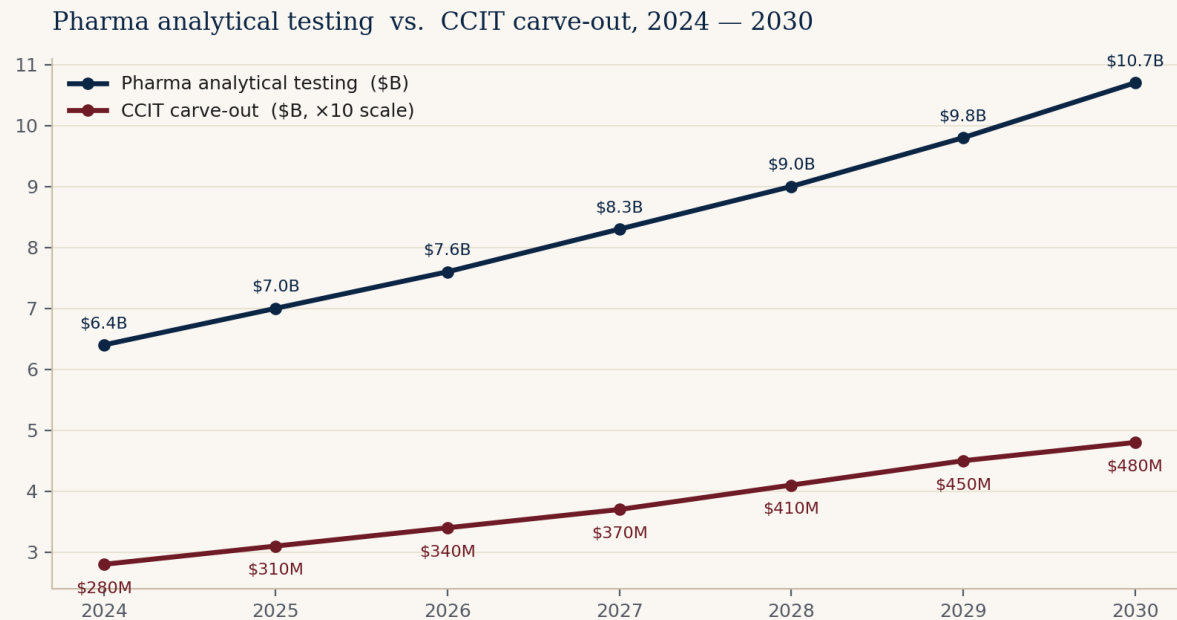
Built to USP <1207>. cGMP from the floor up. Methods filed, not promised. The inspectors already know the address.

Operator's company

Founder-led, scientist-staffed, client-facing PhDs. No hand-offs. The person running your study is the person who designed it.

Where the money sits

A \$6–7 B contract analytical market —
and a \$300 M niche growing twice as fast.



Pharma analytical testing
overall: \$6.4 B, ~9% CAGR.

CCIT carve-out: \$280 M
today → \$475 M by 2030.

That is the bet:
the carve-out grows faster
than the host market — and
CS Analytical is built for the
carve-out, not the host.

Figures are illustrative composites of publicly cited industry ranges. Intended for strategic discussion.

Why now

Four forces are pushing CCIT from afterthought to gating step.

Biologics

Roughly 40% of FDA novel approvals are biologics. Every one of them ships in a sterile primary container.

GLP-1s

Multi-pen, pre-filled, high-volume launches. Integrity defects scale with volume.

Cell, gene, mRNA

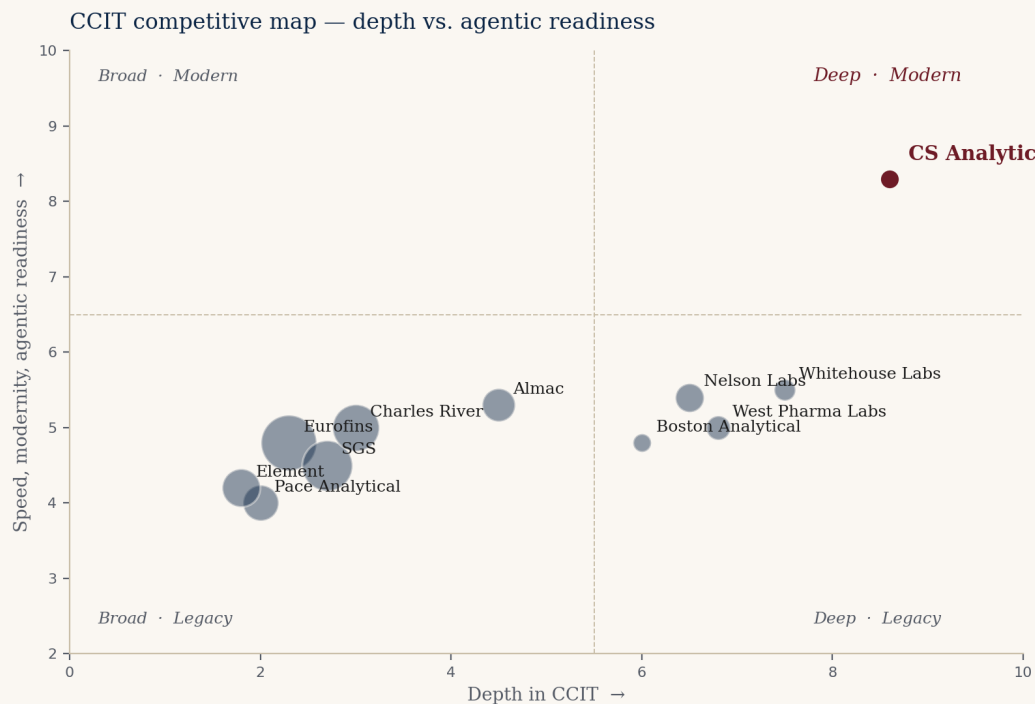
Cryo storage. Closed systems. No room for leak. Methods don't yet exist for half of these formats.

USP <1207>

Deterministic methods are the expectation. Probabilistic methods are increasingly defensive only.

Who is in the room

Big labs are broad. CCIT specialists are few.



The big-five generalists win on footprint and pricing power.

The specialists — Nelson Labs, Whitehouse, Boston Analytical, West — win on depth in one or two modalities.

The upper-right quadrant — deep AND modern AND fast — is essentially empty. That is the CS Analytical position.

Why clients pick us

Vendor-neutral, method-led, founder-accountable.

Vendor-neutral

We own no leak-detection platform — so we recommend the right one. PTI, LDA, Bonfiglioli, ATEQ — we run them all and tell you which fits your container.

Method-led, not box-led

We start with the molecule, the container, and the failure mode. The instrument is the last decision, not the first.

Founder-accountable

Brian signs the study report. There is no account manager between the client and the scientist.

Built to be inspected

Every CS Analytical method is filed with FDA-readiness as the design goal — not a retrofit.

Where we want to go

Five vectors. Sequenced, not stacked.

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- | | | |
|----|-----------------------------------|---|
| 01 | Sterile-injectable CCIT | <i>The core. Defend, deepen, raise price.</i> |
| 02 | Cell & gene cryo containers | <i>New methods, new formats. First mover.</i> |
| 03 | Pre-filled syringe / autoinjector | <i>GLP-1 wave demands repeatable HVLD.</i> |
| 04 | Combination products & device-led | <i>FDA cross-center — few labs are ready.</i> |
| 05 | Regulatory consulting carve-out | <i>Sell the brain, not just the bench.</i> |

Where TSI Citadel multiplies us

We are a 48-person lab. Agentic AI gives us a 480-person reach.

Market radar

Continuous scan of FDA filings, clinical trials, conference rosters, container-vendor catalogs — turned into a weekly BD shortlist.

Regulatory intelligence

Track USP, EP, JP, PIC/S changes in real time. Flag the ones that break a client's filed method.

BD operations

Outbound, qualification, follow-up, meeting prep. The work that adds no science but consumes founder hours.

Scientific literature

Daily distillation of CCIT, HVLD, helium-leak, and container-vendor literature. Methods updates surfaced before the customer asks.

What we'd do first

Three workstreams, ninety days, one readout.

DAYS 1—30

Stand up the agentic stack

TSI Citadel agents wired to FDA, ClinicalTrials.gov, USP, EP, PIC/S, conference rosters, container-vendor release feeds. Weekly BD digest live.

DAYS 31—60

Pilot two growth vectors

Cryo-container method development and autoinjector HVLD. Two named pilot clients each. Methods drafted, quotes out, first revenue booked.

DAYS 61—90

Package and price

Productized CCIT-as-a-service tier. Regulatory-consulting carve-out priced. Public case studies (two) with client co-sign.

What we'd love your help on

An introduction. A read. A reaction.

Introduction

A warm hand-off to the TSI Citadel team — at the level where the agentic-platform partnership conversation can actually begin.

A read

Twenty minutes of your eyes on the long-form market brief. Tell us where the thesis is thin and where it's right.

A reaction

If you see a client, a partner, or a hire we should be in front of — say the name. We'll do the rest.

THANK YOU

The right work,
in the right container,
at the right cadence.

Brian Mulhall

Chief Executive Officer · CS Analytical Laboratory

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